Non Linear Dynamics.

Normal forms. Only one chaotic system

Book: 1) Non Linear Dynamics : Steven Strogatz

MIDSEM
ENDSEM
2 NUMERICAL ASSIGNMENTS

1-D Flow

$$\frac{dx}{dt} = f(t, x(t)) \qquad\qquad \frac{dx}{dt} = f(x(t))$$

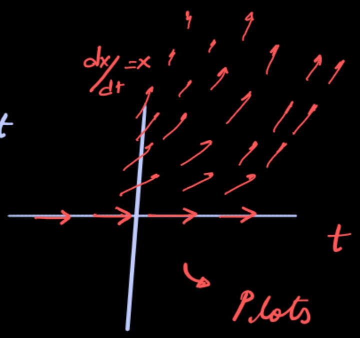
$\downarrow$ Explicit time dependence

$\downarrow$ Implicit, Autonomous systems.

These differential equations along with initial eqns, have an unique solution provided that the function 'f' is continuous.
PICARD'S Theorem

① $\frac{dx}{dt} = x \qquad x(t=0) = x_0 \qquad x(t) = x_0 e^{t}$

↪ Vector fields (quiverplots)



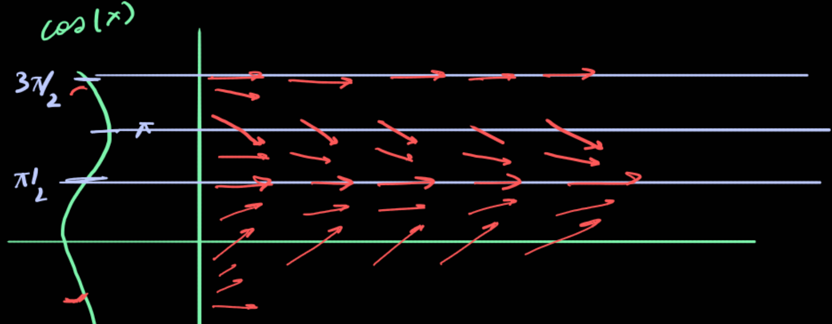
↪ Plots with arrow with slope at (x_0, t_0) is $\left.\frac{dx}{dt}\right|_{(x_0, t_0)}$

② $\frac{dx}{dt} = 1 + x^2 \qquad x(t=0) = x_0$

$$x(t) = \tan\left[t + \arctan(x_0)\right]$$

For times $t^* = \pi/2 - \arctan(x_0) + k\pi$
$x(t) \longrightarrow \infty$

↪ Here the solution goes to ∞ at finite times.
This is called finite escape time.



← flows tend to approach $\pi/2$

$\pi/2$ is a fixed point, which is a stable fixed pt.
$3\pi/2$ is also a fixed pt, which is an unstable one.

The basin of attraction for the fixed pt $\pi/2$ is $(-\pi/2, 3\pi/2)$

<u>Defn</u> [Fixed pts]
Fixed points are points such that $\dot{\vec{x}} = 0 \Rightarrow f(\vec{x}) = 0$ where $\vec{x} \in \mathbb{R}^n$. They are real solutions of $f(\vec{x}) = 0$ where

$$\dot{\vec{x}} = f(\vec{x}).$$

Linear Stability Analysis

$$x(t) = \bar{x} + \epsilon(t)$$

$$\frac{dx(t)}{dt} = \cancel{\frac{d\bar{x}}{dt}}^{\,0} + \frac{d\epsilon}{dt} = f(\bar{x} + \epsilon(t))$$

$$= f(\bar{x}) + f'(\bar{x})\Big|_{\bar{x}} \epsilon + \mathcal{O}(\epsilon^2)$$

$$\Rightarrow \frac{d\epsilon}{dt} = f'(x)\Big|_{\bar{x}} \epsilon$$

$f' > 0 \quad \bar{x} \rightarrow$ unstable
$f' < 0 \quad \rightarrow$ stable

$$\epsilon(t) = e^{t f'(x)} \epsilon(0)$$

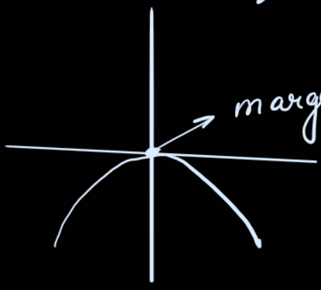
Linear stability analysis only analyses local fixed pts.

Bifurcations

$$\frac{dx}{dt} = \mu - x^2 \qquad f(\vec{x}) = 0 \Rightarrow \vec{x}_{\pm} = \pm\sqrt{\mu}$$

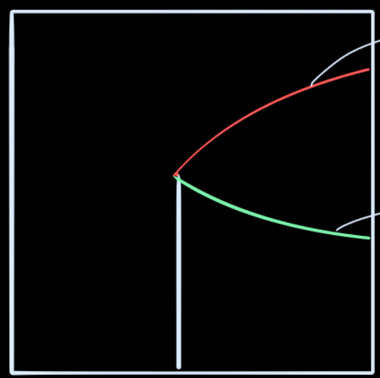
$$f'(x) = \mp 2\sqrt{\mu}$$

Gradient Systems. If $\mu = 0$, then the curve is given



marginally stable.

fixed pts $\vec{x}$



$\sqrt{\mu} \to$ stable pts

$-\sqrt{\mu} \to$ unstable fixed pts

Saddle node Bifurcation.

$\mu = 0$

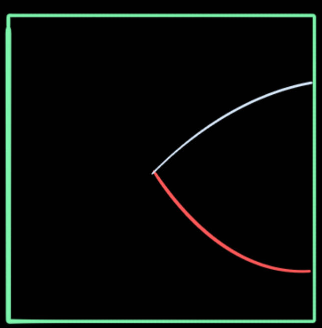
μ → Bifurcation diagram

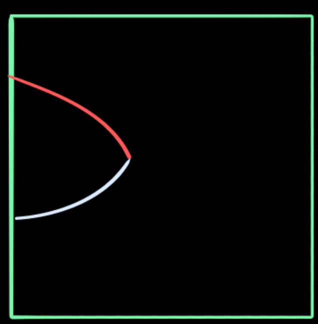
Normal forms → Reducible to $\dot{x} = \mu - x^2$

↳ The bifurcation diagram will depend on the signature of bifurcation.

$\vec{x} \to -\vec{x}$ $\qquad \mu \to -\mu$

$\dot{\vec{x}} = -\mu + x^2$



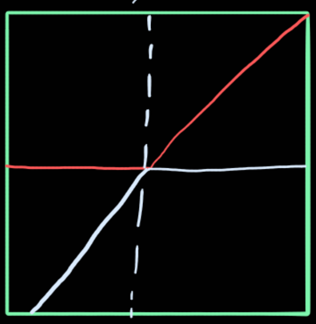


$\dot{x} = \mu x - x^2$ $\qquad f(\vec{x}) = 0$ → $x = 0$, → $x = \mu$ $\qquad \mu$ $f'(x)$ $\quad -\mu$

Transcritical Bifurcation.

$\mu = 0$

$x = 0$



$x = 0$, stable if $\mu < 0$
unstable if $\mu > 0$

$x = \mu$ stable if $\mu > 0$
unstable if $\mu < 0$.

Pitchfork Bifurcation

1D systems $\dot{x} = f(x, \mu)$

N : Population

K : Carrying capacity.

$r > 0$

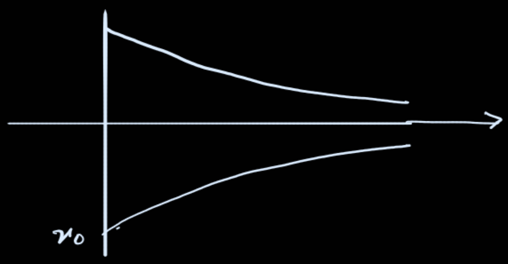
$$\frac{dN}{dt} = rN(1 - N/K)$$

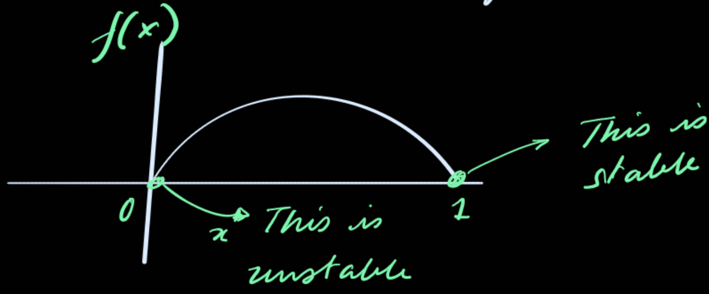
Let $N/K = x$

↳ Population does not grow always and saturates

$$\Rightarrow \frac{dx}{dt} = rx(1-x)$$

→ This is called a logistic model.





$$\left[\begin{array}{l} f'(0) > 0 \\ f'(1) < 0 \rightarrow \text{Phase flows from } 0 \rightarrow 1 \end{array}\right.$$